

(English Medium)

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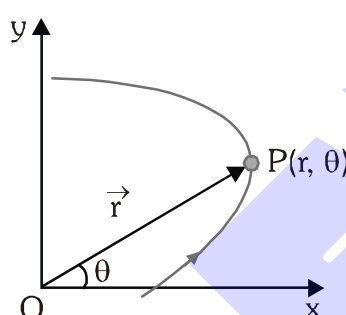
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CIRCULAR MOTION

1.0 DESCRIBING TRANSLATION MOTION BY ANGULAR VARIABLES

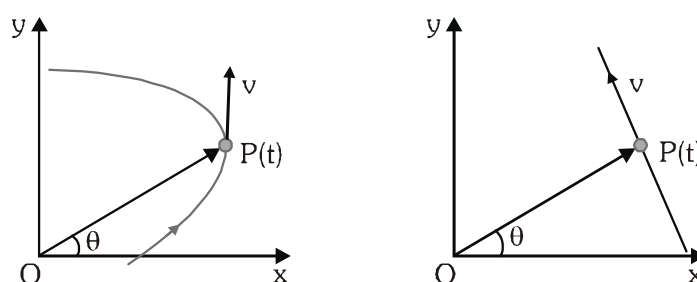
Position of particle can completely be specified by its position vector $\vec{r}$, if magnitude r of the position vector and its orientation relative to some fixed reference direction is known. In the given figure is shown a particle P at location shown by position vector $\vec{r} = \overrightarrow{OP}$. Magnitude of the position vector is distance $r = OP$ of the particle from the origin O and orientation of the position vector is the angle θ made by line OP with the positive x -axis. We now specify position of a particle by these two variables r and θ , known as polar coordinates.



When the particle moves, either or both of these coordinates change with time. If a particle moves radially away from the origin, magnitude r of its position vector $\vec{r}$ increases without any change in angle θ . Similarly, if a particle moves radially towards the origin, r decreases without any change in angle θ . If a particle moves on a circular path with center at the origin, only the angle θ changes with time. If the particle moves on any path other than a radial straight line or circle centered at the origin, both of the coordinates r and θ change with time.

1.1 Angular Motion

Change in direction of position vector $\vec{r}$ is known as angular motion. It happens when a particle moves on a curvilinear path or straight-line path not containing the origin as shown in the following figures.



Angular Motion

Angular position : The coordinate angle θ at an instant is known as angular position of the particle.

Angular Displacement : A change in angular position θ in a time interval is known as angular displacement.

Angular Velocity : The instantaneous rate of change in angular position θ with respect to time is known as angular velocity.

We denote angular velocity by symbol ω . $\omega = \frac{d\theta}{dt}$

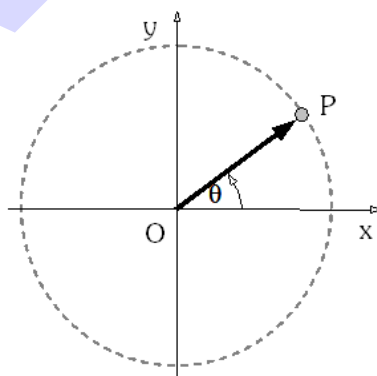
Angular Acceleration : The instantaneous rate of change in angular velocity ω with respect to time is known as angular acceleration.

We denote angular acceleration by symbol α . $\alpha = \frac{d\omega}{dt} = \frac{d^2\theta}{dt^2} = \omega \frac{d\omega}{d\theta}$

If a particle moves in a plane, the position vector turns either in clockwise or anticlockwise sense. Assuming one of these direction positive and other negative, problems of angular motion involving angular position θ , angular velocity ω and angular acceleration α can be solved in fashion similar to problems of rectilinear motion involving position x , velocity v , and acceleration a .

1.2 Kinematics of Circular Motion

A body in circular motion moves on a circular path. At present, we discuss only translation motion of a body on circular path and disregard any rotation; therefore, we represent the body as a particle. In the given figure, a particle P is shown moving on a circular path of radius r . Here only for simplicity center of the circular path is assumed at the origin of a coordinate system. In general, it is not necessary to assume center at the origin. Position vector of the particle is shown by a directed radius $\overline{OP} = \vec{r}$. Therefore, it is also known as radius vector. The radius vector is always normal to the path and has constant magnitude and as the particle moves, it is the angular position θ , which varies with time.



Angular Variables in Circular Motion

Angular position θ , angular velocity ω and angular acceleration α known as angular variables vary in different manner depending on how the particle moves.

Motion with uniform angular velocity

If a particle moves with constant angular velocity, its angular acceleration is zero and position vector turns at constant rate. It is analogous to uniform velocity motion on straight line. The angular position θ at any instant of time t is expressed by the following equation.

$$\theta = \theta_0 + \omega t$$

Motion with uniform angular acceleration

If a particle moves with constant angular acceleration, its angular velocity changes with time at a constant rate. The angular position θ , angular velocity ω and the angular acceleration α bear relations described by the following equations, which have forms similar to corresponding equations that describe uniform acceleration motion.

$$\omega = \omega_0 + \alpha t$$

$$\theta = \theta_0 + \omega_0 t + \frac{1}{2} \alpha t^2$$

$$\theta = \theta_0 + \frac{1}{2} (\omega_0 + \omega) t$$

$$\omega^2 = \omega_0^2 + 2\alpha(\theta - \theta_0)$$

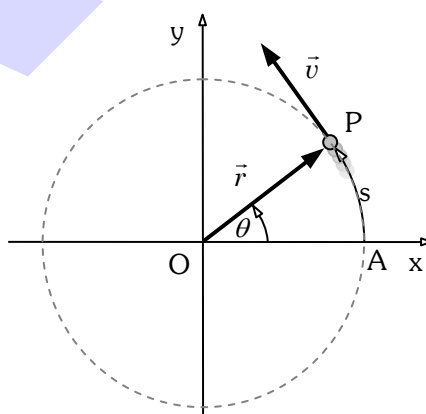
Motion with variable angular acceleration

Variable angular acceleration of a particle is generally specified as function of time, angular position or angular velocity. Problems involving variable angular acceleration can also be solved in a way analogous to corresponding rectilinear motion problems in which acceleration is specified as function of time, position or velocity.

Linear Velocity and Acceleration in Circular Motion

The instantaneous velocity $\vec{v}$ and the instantaneous acceleration $\vec{a}$ are also known as linear velocity and linear acceleration.

In the figure is shown a particle moving on a circular path. As it moves it covers a distance s (arc length).



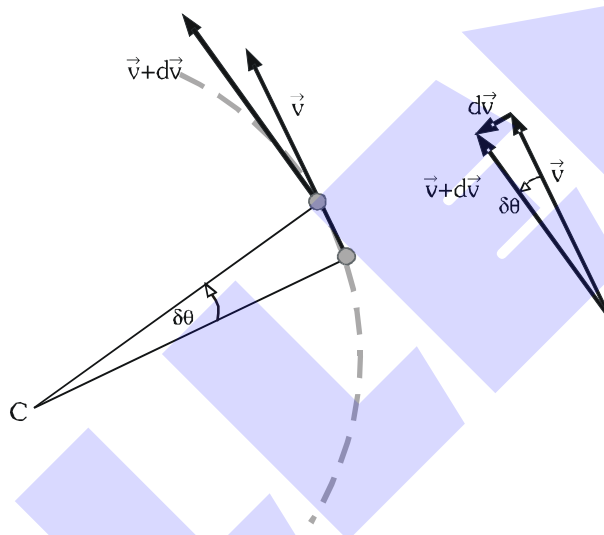
$$s = \theta r$$

Linear velocity $\vec{v}$ is always along the path. Its magnitude known as linear speed is obtained by differentiating s with respect to time t .

$$v = \frac{d\theta}{dt} r = \omega r$$

If speed of the particle is uniform, the circular motion is known as uniform circular motion. In this kind of motion as the particle precedes further, only direction of velocity changes. Therefore, instantaneous acceleration or linear acceleration accounts for only change in direction of motion.

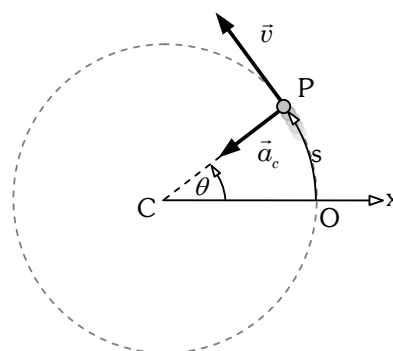
Consider a particle in uniform circular motion. It is shown at two infinitely close instants t and $t + dt$, where its velocity vectors are $\vec{v}$ and $\vec{v} + d\vec{v}$. These two velocity vectors are equal in magnitude and shown in adjacent figure. From this figure, it is obvious that the change $d\vec{v}$ in velocity vector is perpendicular to velocity vector $\vec{v}$ i.e towards the center. It can be approximated as arc of radius equal to magnitude of $\vec{v}$. Therefore we can write $|d\vec{v}| = d\theta \cdot v$. Hence acceleration of this particle is towards the center. It is known as normal component of acceleration or more commonly centripetal acceleration.



Dividing $|d\vec{v}|$ by time interval dt we get magnitude of centripetal acceleration a_c .

$$a_c = \frac{d\theta}{dt} v = \omega v = \omega^2 r = \frac{v^2}{r}$$

Acceleration and velocity of a particle in uniform circular motion are shown in the following figure. To keep the particle in uniform circular motion net force acting on it must be towards the center, therefore it is known as centripetal force. If particle moves with varying speed, the net force on it must have a component along the direction of velocity vector in addition to the centripetal force. This component force is along the tangent to the path and produces a component of acceleration in the tangential direction. This component known as tangential component of acceleration a_T , accounts for change in speed.



$$a_T = \frac{dv}{dt} = \frac{d\omega}{dt} r = \alpha r$$

Illustrations

Illustration 1. Angular position θ of a particle moving on a curvilinear path varies according to the equation $\theta = t^3 - 3t^2 + 4t - 2$, where θ is in radians and time t is in seconds. What is its average angular acceleration in the time interval $t = 2\text{ s}$ to $t = 4\text{ s}$?

Solution Like average linear acceleration, the average angular acceleration α_{av} equals to ratio of change in angular velocity $\Delta\omega$ to the concerned time interval Δt .

$$\alpha_{av} = \frac{\Delta\omega}{\Delta t} = \frac{\omega_{\text{final}} - \omega_{\text{initial}}}{t_{\text{final}} - t_{\text{initial}}} \quad \dots(i)$$

The angular velocity ω being rate of change in angular position can be obtained by equation

$$\omega = \frac{d\theta}{dt}$$

Substituting the given expression of the angular position θ , we have

$$\omega = 3t^2 - 6t + 4 \quad \dots(ii)$$

From the above eq. (ii), angular velocities ω_2 and ω_4 at the given instants $t = 2\text{ s}$ and 4 s are

$$\omega_2 = 4 \text{ rad/s} \quad \text{and} \quad \omega_4 = 28 \text{ rad/s.}$$

Substituting the above values in eq. (1), we have $\alpha_{av} = 12 \text{ rad/s}^2$

Illustration 2. A particle starts from rest and moves on a curve with constant angular acceleration of 3.0 rad/s^2 . An observer starts his stopwatch at a certain instant and record that the particle covers an angular span of 120 rad at the end of 4^{th} second. How long the particle had moved when the observer started his stopwatch?

Solution Let the instants when the particle starts moving and the observer starts his stopwatch, are $t_0 = 0$ to $t = t_1$. Denoting angular positions and angular velocity at the instant $t = t_1$ by θ_1 and ω_1 and the angular position at the instant $t_2 = t_1 + 4\text{ s}$ by θ_2 , we can express the angular span covered during the interval from eq.

$$\theta - \theta_0 = \omega_0 t + \frac{1}{2} \alpha t^2 \rightarrow \theta_2 - \theta_1 = \omega_1 (t_2 - t_1) + \frac{1}{2} \alpha (t_2 - t_1)^2$$

Substituting values θ_1 , θ_2 , t_1 and t_2 , we have $\omega_1 = 24 \text{ rad/s}$

From eq. $\omega = \omega_0 + at$, we have $\omega_1 = \omega_0 + at_1$

Now substituting $\omega_0 = 0$, $\omega_1 = 24$ and $\alpha = 3 \text{ rad/s}^2$, we have $t_1 = 8.0 \text{ s}$

Illustration 3. A particle moves on a circular path of radius 8 m. Distance travelled by the particle in time t is given by the equation $s = \frac{2}{3}t^3$. Find its speed when tangential and normal accelerations have equal magnitude.

Solution The speed v , tangential acceleration a_t and the normal acceleration a_n are expressed by the following equations.

$$v = \frac{ds}{dt}$$

Substituting the given expression for s , we have $v = 2t^2$

$$a_t = \frac{d^2s}{dt^2} \quad \dots(i)$$

Substituting the given expression for s , we have $a_t = 4t$

$$a_n = \frac{v^2}{r} \quad \dots(ii)$$

Substituting v from eq. (i) and $r = 8\text{m}$, we have $a_n = \frac{1}{2}t^4 \quad \dots(iii)$

The instant when the tangential and the normal accelerations have equal magnitude, can be obtained by equating their expressions given in eq. (ii) and (iii).

$$a_n = a_t \rightarrow t = 2\text{s}$$

Substituting the above value of t in eq. (i), we obtain $v = 8\text{ m/s}$

Illustration 4. A particle is moving on a circular path of radius 1.5 m at a constant angular acceleration of 2 rad/s^2 . At the instant $t = 0$, angular speed is $60/\pi\text{ rpm}$. What are its angular speed, angular displacement, linear velocity, tangential acceleration and normal acceleration at the instant $t = 2\text{ s}$.

Solution Initial angular speed is given in rpm (revolution per minute). It is expressed in rad/s as

$$1\text{ rpm} = \frac{2\pi\text{ rad}}{60\text{ s}}$$

$$\omega_0 = \left(\frac{60}{\pi}\right) \times \frac{2\pi\text{ rad}}{60\text{ s}} = 2\text{ rad/s}$$

At the instant $t = 2\text{ s}$, angular speed ω_2 and angular displacement θ_2 are calculated by using eq.

$$\omega_2 = \omega_0 + \alpha t$$

Substituting values $\omega_0 = 2\text{ rad/s}$, $\alpha = 2\text{ rad/s}^2$, $t = 2\text{ s}$, we have

$$\omega_2 = 6\text{ rad/s}$$

$$\theta_2 = \theta_0 + \frac{1}{2}(\omega_0 + \omega_2)t$$

Substituting values $\theta_0 = 0\text{ rad}$, $\omega_0 = 2\text{ rad/s}$, $\omega_2 = 6\text{ rad/s}$ and $t = 2\text{ s}$, we have

$$\theta_2 = 8\text{ rad}$$

Linear velocity at $t = 2$ s, can be calculated by using eq.

$$v_2 = r\omega_2$$

Substituting $r = 1.5$ m and $\omega_2 = 6$ rad/s, we have

$$v_2 = 9 \text{ m/s}$$

Tangential acceleration α_t and normal acceleration α_n can be calculated by using

$$a_t = r\alpha$$

Substituting $r = 1.5$ m and $\alpha = 2$ rad/s², we have

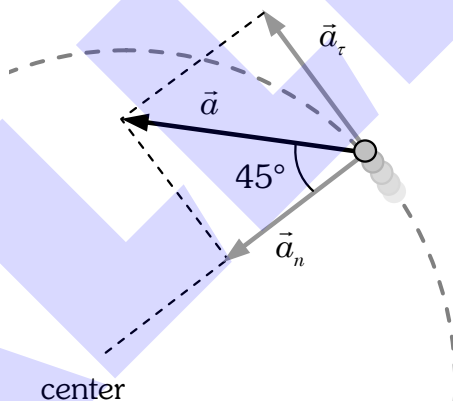
$$a_t = 3 \text{ m/s}^2$$

$$a_n = \omega^2 r$$

Substituting $\omega_2 = 6$ rad/s and $r = 1.5$ m, we have

$$a_n = 54 \text{ m/s}^2$$

Illustration 5. A particle is moving in a circular orbit with a constant tangential acceleration. After 2 s from the beginning of motion, angle between the total acceleration vector and the radius R becomes 45° . What is the angular acceleration of the particle?



Solution

In the adjoining figure are shown the total acceleration vector $\vec{a}$ and its components the tangential accelerations $\vec{a}_t$ and normal accelerations $\vec{a}_n$ are shown. These two components are always mutually perpendicular to each other and act along the tangent to the circle and radius respectively. Therefore, if the total acceleration vector makes an angle of 45° with the radius, both the tangential and the normal components must be equal in magnitude. Now from eq. and , we have

$$a_t = a_n \rightarrow \alpha R = \omega^2 R \Rightarrow \alpha = \omega^2 \quad \dots(i)$$

Since angular acceleration is uniform, from eq., we have $\omega = \omega_0 + \alpha t$

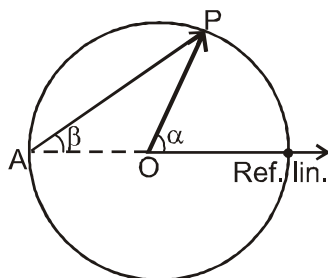
Substituting $\omega_0 = 0$ and $t = 2$ s, we have $\omega = 2\alpha \dots(ii)$

From eq. (i) and (ii), we have $\alpha = 0.25 \text{ rad/s}^2$

2.0 ANGULAR VELOCITY OF A PARTICLE RELATIVE TO ANOTHER PARTICLE OR POINT

Just as velocities are always relative, similarly angular velocity is also always relative. There is no such thing as absolute angular velocity. Angular velocity is defined with respect to origin, the point from which the position vector of the moving particle is drawn.

Consider a particle P moving along a circular path shown in the figure given below



Here angular velocity of the particle P w.r.t. 'O' and 'A' will be different

Angular velocity of a particle P w.r.t. O, $\omega_{PO} = \frac{d\alpha}{dt}$

Angular velocity of a particle P w.r.t. A, $\omega_{PA} = \frac{d\beta}{dt}$

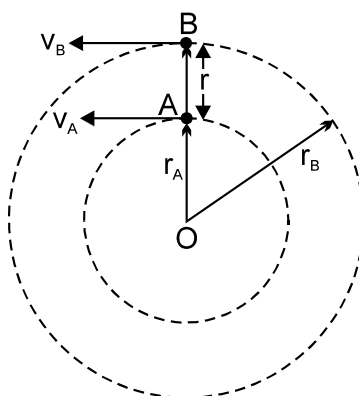
Definition :

Angular velocity of a particle 'A' with respect to the other moving particle 'B' is the rate at which position vector of 'A' with respect to 'B' rotates at that instant. (or it is simply, angular velocity of A with origin fixed at B). Angular velocity of A w.r.t. B, ω_{AB} is mathematically define as

$$\omega_{AB} = \frac{\text{Component of relative velocity of A w.r.t. B, perpendicular to line}}{\text{separation between A and B}} = \frac{(V_{AB})_{\perp}}{r_{AB}}$$

Important points :

- If two particles are moving on two different concentric circles with different velocities then angular velocity of B as observed by A will depend on their positions and velocities. Consider the case when A and B are closest to each other moving in same direction as shown in figure. In this situation



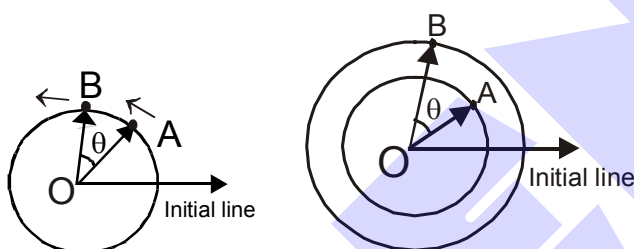
$$(V_{AB})_{\perp} = v_B - v_A$$

Separation between A and B is $r_{BA} = r_B - r_A$

$$\text{so, } \omega_{AB} = \frac{(V_{AB})_{\perp}}{r_{AB}} = \frac{v_B - v_A}{r_B - r_A}$$

- If two particles are moving on the same circle or different coplanar concentric circles in same direction with different uniform angular speed ω_A and ω_B respectively, the rate of change of angle between $\overline{OA}$ and $\overline{OB}$ is

$$\frac{d\theta}{dt} = \omega_B - \omega_A$$



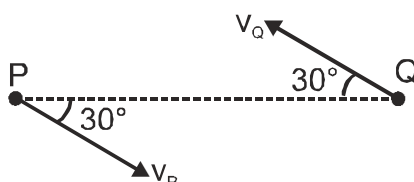
So the time taken by one to complete one revolution around O w.r.t. the other

$$T = \frac{2\pi}{\omega_{\text{rel}}} = \frac{2\pi}{\omega_2 - \omega_1} = \frac{T_1 T_2}{T_1 - T_2}$$

- $\omega_B - \omega_A$ is rate of change of angle between $\overline{OA}$ and $\overline{OB}$. This is not angular velocity of B w.r.t. A. (Which is rate at which line AB rotates)

Illustrations

Illustration 6. Two moving particles P and Q are 10 m apart at any instant. Velocity of P is 8 m/s at 30° , from line joining the P and Q and velocity of Q is 6 m/s at 30° . Calculate the angular velocity of P w.r.t. Q



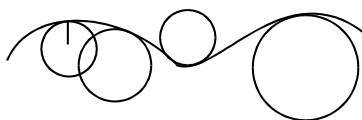
Solution $\omega_{PQ} = \frac{8 \sin 30^\circ - (-6 \sin 30^\circ)}{10} = 0.7 \text{ rad/s.}$

3.0 RADIUS OF CURVATURE

Any curved path can be assumed to be made of infinite circular arcs. Radius of curvature at a point is the radius of the circular arc at a particular point which fits the curve at that point.

$$F_c = \frac{mv^2}{R} \Rightarrow R = \frac{mv^2}{F_c} = \frac{mv^2}{F_\perp}$$

$F_\perp$ = Force perpendicular to velocity (centripetal force)



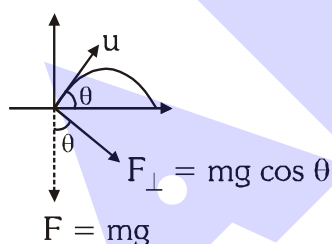
If the equation of trajectory of a particle is given we can find the radius of curvature of the instantaneous circle by using the formula,

$$R = \frac{\left[1 + \left(\frac{dy}{dx}\right)^2\right]^{3/2}}{\left|\frac{d^2y}{dx^2}\right|}$$

Illustrations

Illustration 7. A particle of mass m is projected with speed u at an angle θ with the horizontal. Find the radius of curvature of the path traced out by the particle at the point of projection and also at the highest point of trajectory.

Solution



At point of projection

$$R = \frac{mv^2}{F_\perp} = \frac{mu^2}{mg \cos \theta}$$

$$R = \frac{u^2}{g \cos \theta} \quad \text{Ans.}$$

at highest point

$$a_\perp = g, \quad v = u \cos \theta : R = \frac{v^2}{a_\perp} = \frac{u^2 \cos^2 \theta}{g}$$

Illustration 8. A particle moves along the plane trajectory $y(x)$ with constant speed v . Find the radius of curvature of the trajectory at the point $x = 0$ if the trajectory has the form of a parabola $y = ax^2$ where 'a' is a positive constant.

Solution If the equation of the trajectory of a particle is given we can find the radius of trajectory of the instantaneous circle by using the formula

$$R = \frac{\left[1 + \left(\frac{dy}{dx}\right)^2\right]^{3/2}}{\left|\frac{d^2y}{dx^2}\right|}$$

$$\text{As; } y = ax^2 \Rightarrow \frac{dy}{dx} = 2ax = 0 \text{ (at } x = 0) \text{ and } \frac{d^2y}{dx^2} = 2a$$

Now radius of trajectory is given by

$$R = \frac{[1 + 0]^{3/2}}{2a} = \frac{1}{2a}$$

Alter : This problem can also be solved by using the formula: $R = \frac{v^2}{a_{\perp}}$, $y = ax^2$,

$$\text{differentiate with respect to time } \frac{dy}{dt} = 2ax \frac{dx}{dt} \dots (1)$$

$$\text{at } x = 0, v_y = \frac{dy}{dt} = 0 \text{ hence } v_x = v$$

$$\text{Now, differentiate (1) with respect to time } \frac{d^2y}{dt^2} = 2ax \frac{d^2x}{dt^2} + 2a \left(\frac{dx}{dt}\right)^2$$

$$\text{at } x = 0, v_x = v$$

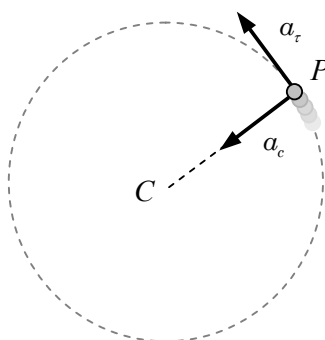
$\therefore$ net acceleration, $a = a_y = 2av^2$ (since $ax = 0$) this acceleration is perpendicular to velocity (v_x).

Hence it is equal to centripetal acceleration

$$R = \frac{v^2}{a_{\perp}} = \frac{v^2}{2av^2} = \frac{1}{2a} \text{ Ans.}$$

4.0 DYNAMICS OF CIRCULAR MOTION

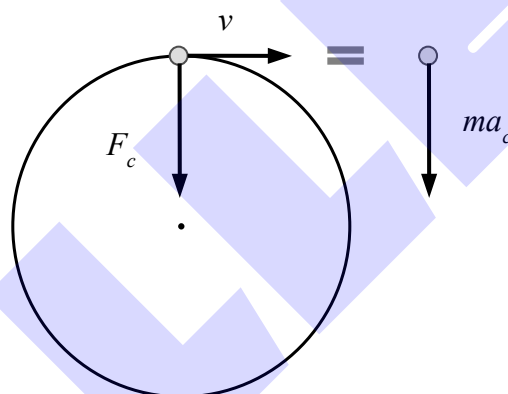
Velocity vector points always tangent to the path and continuously change its direction, as a particle moves on a circular path even with constant speed and give rise to normal component of acceleration, which always points toward the center of the circular path. This component of acceleration is known as centripetal (center seeking) acceleration and denoted by a_c . Moreover, if speed also changes the particle will have an additional acceleration component along the tangent to the path. This component of acceleration is known as tangential acceleration and denoted by a_t .



The centripetal and tangential accelerations

Application of Newton's Second law in Circular Motion

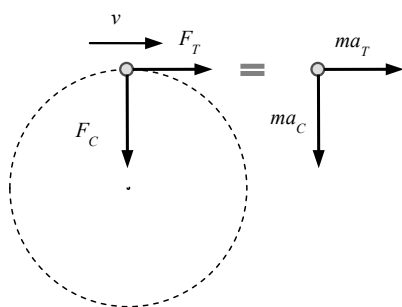
Consider a particle of mass m moving with uniform speed v in a circle of radius r as shown in figure. It necessarily possess a centripetal acceleration and hence there must be a net force ($\sum \vec{F} = \vec{F}_C$) acting always towards the center according to the second law. This net force F_C acting towards the center is known as centripetal force.



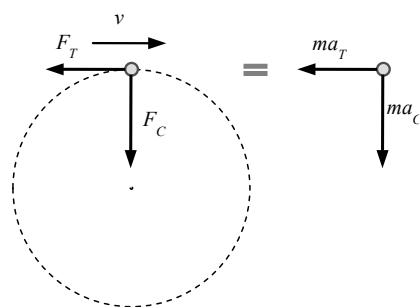
$$\sum \vec{F} = m\vec{a} \rightarrow F_C = ma_C$$

When a particle is whirled with the help of a string in a horizontal circle, the required centripetal force is the tension in the string. The gravitational attraction between a satellite and the earth, between moon and the earth, between the sun and its planets, and the electrostatic attraction between the nucleus and electrons are the centripetal forces and provide the necessary centripetal acceleration.

Now consider a particle moving on circular path with varying speed. The net acceleration has two components, the tangential acceleration and the centripetal acceleration. Therefore, the net force must also have two components, one component in tangential direction to provide the tangential acceleration and the other component towards the center to provide the centripetal acceleration. The former one is known as tangential force and the latter one as centripetal force.



Particle moving with increasing speed.



Particle moving with decreasing speed.

When the particle moves with increasing speed the tangential force acts in the direction of motion and when the particle moves with decreasing speed the tangential force acts in direction opposite to direction of motion.

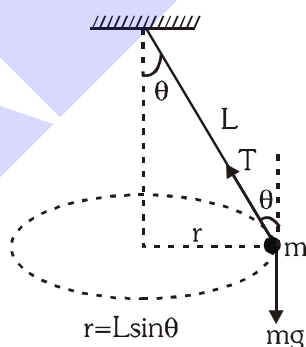
To write equations according to the second law, we consider the tangential and the radial directions as two mutually perpendicular axes. The components along the tangential and the radial directions are designated by subscripts T and C.

$$F_C = ma_C$$

$$F_T = ma_T$$

4.1 Conical Pendulum

If a small particle of mass m tied to a string is whirled in a horizontal circle, as shown in figure. The arrangement is called the 'conical pendulum'. In case of conical pendulum the vertical component of tension balances the weight while its horizontal component provides the necessary centripetal force. Thus,,



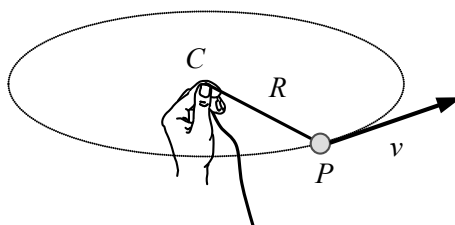
$$T \sin \theta = \frac{mv^2}{r} \text{ and } T \cos \theta = mg \Rightarrow v = \sqrt{rg \tan \theta}$$

$$\therefore \text{Angular speed } \omega = \frac{v}{r} = \sqrt{\frac{g \tan \theta}{r}}$$

$$\text{So, the time period of pendulum is } T = \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{r}{g \tan \theta}} = 2\pi \sqrt{\frac{L \cos \theta}{g}}$$

Illustrations

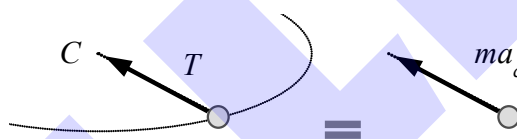
Illustration 9. In free space, a man whirls a small stone P of mass m with the help of a light string in a circle of radius R as shown in the figure. Establish the relation between the speed of the stone and the tension developed in the string. Also, find the force applied on the string by the man.

**Solution**

The system is in free space therefore no force other than the tension acts on the stone to provide necessary centripetal force. The tension does not have any component in tangential direction therefore tangential component of acceleration is zero. In the adjoining figure it is shown that how tension (T) in the string produces necessary centripetal force.

Applying Newton's second law of motion to the stone,

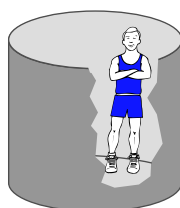
$$\text{we have } F_c = ma_c \rightarrow T = ma_c = \frac{mv^2}{R}$$



The end where man holds the string is stationary and tension applied by the string to this end is T towards the stone, therefore the man must apply a force equal to the tension in magnitude but in a direction away from the stone.

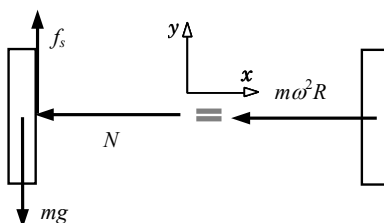
$$T = \frac{mv^2}{R}$$

Illustration 10. A boy stands on a horizontal platform inside a cylindrical container of radius R resting his back on the inner surface of the container. The container can be rotated about the vertical axis of symmetry. The coefficient of static friction between his back and the inner surface of the container is μ_s . The angular speed of the container is gradually increased. Find the minimum angular speed at which if the platform below his feet removed, the boy should not fall.



Solution

As the container rotates at angular speed ω the boy moves in a circular path of radius R with a speed $v = \omega R$. Since the angular speed is increased gradually the angular acceleration can be ignored and hence the tangential acceleration of the boy too. Thus, the boy has a centripetal acceleration of $\omega^2 R$, provided by the normal reaction N applied by the wall of the container. The weight of the boy is balanced by the force of static friction. All these force are shown in the adjoining figure where the boy is shown schematically by a rectangular box of mass m .



$$\sum F_x = ma_x \rightarrow N = m\omega^2 R \quad \dots(i)$$

$$\sum F_y = ma_y \rightarrow f_s = mg \quad \dots(ii)$$

Since force of static friction cannot be greater than the limiting friction $\mu_s N$,

$$\text{we have } f_s \leq \mu_s N \quad \dots(iii)$$

From the above equations, the minimum angular speed is $\omega_{\min} = \sqrt{\frac{g}{\mu_s R}}$

5.0 CIRCULAR TURNING ON ROADS

When vehicles go through turnings, they travel along a nearly circular arc. There must be some force which will produce the required centripetal acceleration. If the vehicles travel in a horizontal circular path, this resultant force is also horizontal. The necessary centripetal force is being provided to the vehicles by following three ways.

1. **By friction only.**
2. **By banking of roads only.**
3. **By friction and banking of roads both.**

In real life the necessary centripetal force is provided by friction and banking of roads both. Now let us write equations of motion in each of the three cases separately and see what are the constant in each case.

By Friction Only

Suppose a car of mass m is moving at a speed v in a horizontal circular arc of radius r . In this case, the necessary centripetal force to the car will be provided by force of friction f acting towards center

$$\text{Thus, } f = \frac{mv^2}{r}$$

Further, limiting value of f is μN or $f_L = \mu N = \mu mg$ ($N = mg$)

Therefore, for a safe turn without sliding $\frac{mv^2}{r} \leq f_L$ or $\frac{mv^2}{r} \leq \mu mg$ or $\mu \geq \frac{v^2}{rg}$ or $v \leq \sqrt{\mu rg}$

Here, two situations may arise. If μ and r are known to us, the speed of the vehicle should not exceed $\sqrt{\mu rg}$ and if v and r are known to us, the coefficient of friction should be greater than $\frac{v^2}{rg}$

By Banking of Roads Only

Friction is not always reliable at circular turns if high speeds and sharp turns are involved to avoid dependence on friction, the roads are banked at the turn so that the outer part of the road is some what lifted compared to the inner part.

Applying Newton's second law along the radial direction

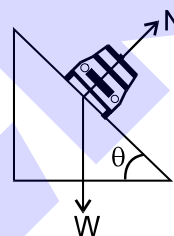
$$N \sin \theta = \frac{mv^2}{r}$$

Net force along vertical direction should be zero

or $N \cos \theta = mg$

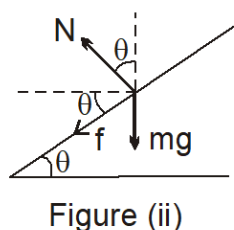
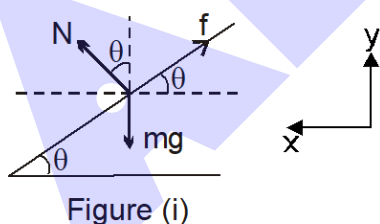
from these two equations, we get

$$\tan \theta = \frac{v^2}{rg} \text{ or } v = \sqrt{rg \tan \theta}$$



By Friction and Banking of Road Both

If a vehicle is moving on a circular road which is rough and banked also, then three forces may act on the vehicle, of these the first force, i.e., weight (mg) is fixed both in magnitude and direction.



The direction of second force, i.e., normal reaction N is also fixed (perpendicular to road) while the direction of the third force i.e., friction f can be either inwards or outwards while its magnitude can be varied upto a maximum limit ($f_L = \mu N$). So the magnitude of normal reaction N and directions plus magnitude of friction f are so adjusted that the resultant of the three forces mentioned above is $\frac{mv^2}{r}$ towards the center. Of these m and r are also constant. Therefore, magnitude of N and directions plus magnitude of friction mainly depends on the speed of the vehicle v . Thus, situation varies from problem to problem. Even though we can see that :

- (i) Friction f will be outwards if the vehicle is at rest $v = 0$. Because in that case the component of weight $mg \sin \theta$ is balanced by f .
- (ii) Friction f will be inwards if $v > \sqrt{rg \tan \theta}$
- (iii) Friction f will be outwards if $v < \sqrt{rg \tan \theta}$ and
- (iv) Friction f will be zero if $v = \sqrt{rg \tan \theta}$
- (v) For maximum safe speed (figure (ii))

$$N \sin \theta + f \cos \theta = \frac{mv^2}{r} \quad \text{.....(i)}$$

$$N \cos \theta - f \sin \theta = mg \quad \text{.....(ii)}$$

As maximum value of friction $f = \mu N$

$$\therefore \frac{\sin \theta + \mu \cos \theta}{\cos \theta - \mu \sin \theta} = \frac{v^2}{rg} \quad \therefore v_{\max} = \sqrt{\frac{rg(\tan \theta + \mu)}{(1 - \mu \tan \theta)}}$$

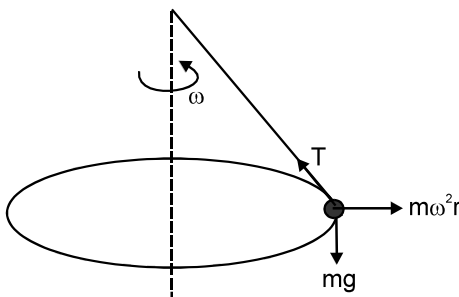
Similarly for minimum safe speed (see figure (i)); $v_{\min} = \sqrt{\frac{rg(\tan \theta - \mu)}{(1 + \mu \tan \theta)}}$

6.0 CENTRIFUGAL FORCE

When a body is rotating in a circular path and the centripetal force vanishes, the body would leave the circular path. To an observer A who is not sharing the motion along the circular path, the body appears to fly off tangentially at the point of release. To another observer B, who is sharing the motion along the circular path (i.e., the observer B is also rotating with the body which is released, it appears to B, as if it has been thrown off along the radius away from the centre by some force. This inertial force is called centrifugal force.)

Its magnitude is equal to that of the centripetal force. $= \frac{mv^2}{r} = m\omega^2 r$

Direction of centrifugal force, it is always directed radially outward. Centrifugal force is a fictitious force which has to be applied as a concept only in a rotating frame of reference to apply Newton's law of motion in that frame. FBD of ball w.r.t. non inertial frame rotating with the ball. Suppose we are working from a frame of reference that is rotating at a constant, angular velocity ω with respect to an inertial frame. If we analyse the dynamics of a particle of mass m kept at a distance r from the axis of rotation, we have to assume that a force $m\omega^2 r$ react radially outward on the particle. Only then we can apply Newton's laws of motion in the rotating frame. This radially outward pseudo force is called the centrifugal force.



Illustrations

Illustration 11. Find the maximum speed at which a car can turn round a curve of 30 m radius on a level road if the coefficient of friction between the tyres and the road is 0.4 [acceleration due to gravity = 10 m/s²]

Solution Here centripetal force is provided by friction so

$$\frac{mv^2}{r} \leq \mu mg \Rightarrow v_{\max} = \sqrt{\mu rg} = \sqrt{120} \approx 11 \text{ ms}^{-1}$$

Illustration 12. For traffic moving at 60 km/hr, if the radius of the curve is 0.1 km, what is the correct angle of banking of the road ? ($g = 10 \text{ m/s}^2$)

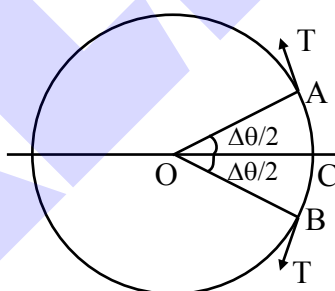
Solution In case of banking $\tan \theta = \frac{v^2}{rg}$

$$\text{Here } v = 60 \times \frac{5}{18} \text{ ms}^{-1} = \frac{50}{3} \text{ ms}^{-1} \quad r = 0.1 \text{ km} = 100 \text{ m}$$

$$\text{So } \tan \theta = \frac{50/3 \times 50/3}{100 \times 10} = \frac{5}{18} \Rightarrow \theta = \tan^{-1} \left(\frac{5}{18} \right)$$

Illustration 13. A metal ring of mass m and radius R is placed on a smooth horizontal table and is set rotating about its own axis in such a way that each part of the ring moves with a speed v . Find the tension in the ring.

Solution Consider a small part ACB of the ring that subtends an angle $\Delta\theta$ at the centre as shown in figure. Let the tension in the ring be T .



The forces on this small part ACB are

- (a) tension T by the part of the ring left to A,
- (b) tension T by the part of the ring right to B,
- (c) weight $(\Delta m)g$ and
- (d) normal force N by the table.

The tension at A acts along the tangent at A and the tension at B acts along the tangent at B. As the small part ACB moves in a circle of radius R at a constant speed v , its acceleration is towards the centre (along CO) and has a magnitude $(\Delta m)v^2/R$.

Resolving the forces along the radius, CO,

$$T \cos\left(90^\circ - \frac{\Delta\theta}{2}\right) + T \cos\left(90^\circ - \frac{\Delta\theta}{2}\right) = (\Delta m) \left(\frac{v^2}{R}\right)$$

$$\text{Or, } 2T \sin \frac{\Delta\theta}{2} = (\Delta m) \left(\frac{v^2}{R}\right) \quad \dots (i)$$

The length of the part ACB is $R\Delta\theta$. As the total mass of the ring is m , the mass of the part ACB will be

$$\Delta m = \frac{m}{2\pi R} R\Delta\theta = \frac{m\Delta\theta}{2\pi}$$

Putting Δm in (i),

$$2T \sin \frac{\Delta\theta}{2} = \frac{m}{2\pi} \Delta\theta \left(\frac{v^2}{R}\right)$$

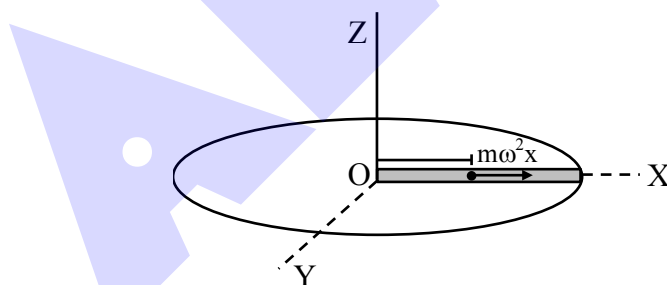
$$\text{Or, } T = \frac{mv^2}{2\pi R} \frac{\Delta\theta/2}{\sin(\Delta\theta/2)}$$

$$\text{As } \Delta\theta \text{ is very small, } \frac{\Delta\theta/2}{\sin(\Delta\theta/2)} \approx 1 \text{ and } T = \frac{mv^2}{2\pi R}$$

Illustration 14. A table with smooth horizontal surface is turning at an angular speed ω about its axis. A groove is made on the surface along a radius and a particle is gently placed inside the groove at a distance a from the centre. Find the speed of the particle as its distance from the centre becomes L .

Solution

The situation is shown in figure.



Let us work from the frame of reference of the table. Let us take the origin at the centre of rotation O and the X -axis along the groove (figure). The Y -axis is along the line perpendicular to OX coplanar with the surface of the table and the Z -axis is along the vertical. Suppose at time t the particle in the groove is at a distance x from the origin and is moving along the X -axis with a speed v . The forces acting on the particle (including the pseudo forces that we must assume because we have taken our frame on the table which is rotating and is noninertial) are

- (a) weight mg vertically downward,
- (b) normal contact force N_1 vertically upward by the bottom surface of the groove,

- (c) normal contact force N_2 parallel to the Y-axis by the side walls of the groove,
- (d) centrifugal force $m\omega^2 x$ along the X-axis, and
- (e) coriolis force along Y-axis (coriolis force is perpendicular to the velocity of the particle and the axis of rotation.)

As the particle can only move in the groove, its acceleration is along the X-axis. The only force along the X-axis is the centrifugal force $m\omega^2 x$. All the other forces are perpendicular to the X-axis and have no components along the X-axis.

Thus, the acceleration along the X-axis is

$$a = \frac{F}{m} = \frac{m\omega^2 x}{m} = \omega^2 x$$

$$\text{or, } \frac{dv}{dt} = \omega^2 x$$

$$\text{or, } \frac{dv}{dt} \cdot \frac{dx}{dt} = \omega^2 x$$

$$\text{or, } \frac{dv}{dt} \cdot v = \omega^2 x$$

$$\text{or, } v dv = \omega^2 x dx$$

$$\text{or, } \int_0^v v dv = \int_a^L \omega^2 x dx$$

$$\text{or, } \left[\frac{1}{2} v^2 \right]_0^v = \left[\frac{1}{2} \omega^2 x^2 \right]_a^L$$

$$\text{or, } \frac{v^2}{2} = \frac{1}{2} \omega^2 (L^2 - a^2)$$

$$\text{or, } v = \omega \sqrt{L^2 - a^2}$$